Numerical processing of financial data - Session 5

6/02/2025

▷ Back to the option hedging problem

The setting:

market
- Asset S_t (single stock, stock index SP500, other)
- interest rate $r > 0$

new
- Option payoff $g(S_T)$ to be hedged (point of view of the option seller ↳ an investment bank)

▷ option maturity : T (1 month, 6 months)

- Self-financing portfolio

$$V_t = \underbrace{\delta_t S_t}_{\substack{\text{delta port} \\ \text{(risky port)}}} + \underbrace{\delta_t^0 S_t^0}_{\substack{\text{cash account} \\ \text{(no risks)}}}$$

- <u>Dynamics of V</u>: self-financing

$$V_{t+\Delta t} - V_t = \delta_t \left(S_{t+\Delta t} - S_t \right) + r \underbrace{\left(V_t - \delta_t S_t \right)}_{\text{cash part}} \Delta t$$

(for example $\Delta t = 1$ day, 1 week, ...)

$$= \delta_t \underbrace{\Delta S_t}_{\text{random variation}} + r \left(V_t - \delta_t S_t \right) \Delta t$$

$\downarrow$ deterministic evolution with rate r

where $\Delta S_t = S_{t+\Delta t} - S_t$

- Total position of the option seller:

$$- \text{option} + \text{portfolio}$$

$$- \underbrace{P(t, S_t)} + V_t$$

function $P(t, S)$ we have to characterise

▷ PnL of the option hedger between t and $t+\Delta t$:

$$\begin{aligned}
PnL_{t,t+\Delta t} &= (\text{Value at } t+\Delta t) - (\text{Value at } t) \\
&= \left(-P(t+\Delta t, S_{t+\Delta t}) + V_{t+\Delta t}\right) \\
&\quad - \left(-P(t, S_t) + V_t\right) \\
&= -\left(P(t+\Delta t, S_{t+\Delta t}) - P(t, S_t)\right) \\
&\quad + V_{t+\Delta t} - V_t \\
&= \Delta(\text{option}) + \Delta V
\end{aligned}$$

- <u>We ask</u> that the function P is $C^{1,2}$

$$(t,s) \mapsto P(t,s) \in C^{1,2}([0,T) \times [0,\infty))$$

$\hookrightarrow$ means $\partial_t P(t,s)$, $\partial_s P(t,s)$, $\partial_{ss} P(t,s)$ exist and are continuous

- We can develop the term ΔP using a Taylor development

$$PnL_{t,t+\Delta t} = -(P(t+\Delta t, S_{t+\Delta t}) - P(t,S_t)) + V_{t+\Delta t} - V_t$$

$$= -\Big[\partial_t P(t,S_t)\Delta t + \cancel{\partial_s P(t,S_t)\Delta S_t} + \frac{1}{2}\partial_{ss} P(t,S_t)(\Delta S_t)^2 + \underbrace{o(\Delta t) + o((\Delta S_t)^2)}_{\text{remainder terms}}\Big]$$

$$+ \cancel{\delta_t \Delta S_t} + r(V_t - \delta_t S_t)\Delta t$$

If we make the choice

$$\boxed{\delta_t = \partial_s P(t,S_t)}$$

then, we remove the linear term in ΔS_t from the PnL

$\hookrightarrow$ this is precisely what delta-hedging does

$\hookrightarrow$ total position is "delta neutral"

<u>Reminder:</u> $f(x) = x(x)$ as $x \to 0$

means $\limsup_{x \to 0} \frac{|f(x)|}{x} = 0$

We get

$$\mathrm{PnL}_{t,t+\Delta t} = -\Big[\partial_t P(t,S_t) - rV_t + r\underbrace{\partial_S P(t,S_t)S_t}_{\text{or } \delta_t}\Big]\Delta t$$
$$- \frac{1}{2}\partial_{SS}P(t,S_t)(\Delta S_t)^2$$
$$+ o(\Delta t) + o((\Delta S_t)^2)$$

We <u>decide</u> to keep terms up to order 1 in Δt

up to order 2 in ΔS_t

↳ this corresponds to assuming that, on average

$$\overline{\Delta S_t^2} \approx c \cdot \Delta t$$

$$|\Delta S_t| \approx \text{const} \cdot \sqrt{\Delta t}$$

↳ Typical situation when S behaves as a Brownian motion

↳ On financial market, this approximation is appropriate in low frequency/mid frequency setting

$\Delta t \approx 1$ day, $\Delta t \approx 1$ h

Let us denote

$$\Theta(t,S_t) = \partial_t P(t,S_t) + r\,\partial_S P(t,S_t) S_t - r P(t,S_t)$$

we have

$$\boxed{\begin{aligned} \mathrm{PnL}_{t,t+\Delta t} = \; & r\big(V_t - P(t,S_t)\big)\Delta t - \Theta(t,S_t)\Delta t \\ & + \frac{1}{2}\partial_{SS} P(t,S_t)\,\Delta S_t^2 + o(\Delta t) + o\big((\Delta S_t)^2\big) \end{aligned}}$$

- Let us consider the case $\underline{r=0}$ so to simplify

In this case, $\Theta(t,S_t) = \partial_t P(t,S_t)$, and

$$\boxed{\mathrm{PnL}_{t,t+\Delta t} \underset{\uparrow}{\approx} -\partial_t P(t,S_t)\Delta t - \frac{1}{2}\partial_{SS} P(t,S_t)(\Delta S_t)^2}$$

at order 1 in Δt and 2 in (ΔS_t)

- Comments:

▷ If the function P is such that

$$\partial_t P(t,S) > 0 \quad \text{and} \quad \partial_{SS} P(t,S) > 0 \qquad \forall t \quad \forall S > 0$$

then, we would have

$$\mathrm{PnL}_{t,t+\Delta} < 0 \qquad \text{always}$$

(at order 1 in Δt, order 2 in ΔS_t)

↳ we would systematically loose money

▷ If, on the contrary,

$$\partial_t P(t,S) < 0 \quad \text{and} \quad \partial_{SS} P(t,S) < 0 \qquad \forall t \quad \forall S > 0$$

then $\quad \mathrm{PnL}_{t,t+\Delta t} > 0 \quad$ always

Possible? We would have

$$\mathrm{PnL}_{0,T} = \sum_{i=0}^{n-1} \underbrace{\mathrm{PnL}_{t_i,t_{i+1}}}_{>0} > 0 \quad \text{always}$$

$\longrightarrow$ this would be an arbitrage.

(We would not be able to sell the option for the price $P(0,S_0)$)

Consequence: the two functions $\Theta(t,S) = \partial_t P(t,S)$

and $\partial_{SS} P(t,S)$

cannot have the same sign $\forall t, \forall S$.

▷ <u>Our criterion</u>: we impose that the approximate PnL is zero <u>on average</u>

$$E[\,PnL_{t,t+\Delta t} \mid \mathcal{F}_t\,] = 0$$

market information at time t : observations of S up to t

$\mathcal{F}_t = \sigma(S_u, u \le t)$

the σ-algebra generated by S up to time t

With this condition, we get:

$$E[\,PnL_{t,t+\Delta t} \mid \mathcal{F}_t\,] = 0 \Rightarrow$$

$$-\partial_t P(t,S_t)\Delta t - \frac{1}{2}\partial_{ss}P(t,S_t)S_t^2 \underbrace{E\left[\left(\frac{\Delta S_t}{S_t}\right)^2 \mid \mathcal{F}_t\right]}_{E[(R_{t,t+\Delta t})^2 \mid \mathcal{F}_t]} = 0$$

where $R_{t,t+\Delta t} = \dfrac{S_{t+\Delta t} - S_t}{S_t}$ = (relative) return of S over $[t, t+\Delta t]$

$(R_{t,t+\Delta t})^2$: a measure of the <u>squared</u> <u>realized</u> <u>volatility</u> of S over $(t, t+\Delta t)$

$E[(R_{t,t+\Delta t})^2 \mid \mathcal{F}_t]$ best prediction of the squared realized volatility of S over the time interval $[t, t+\Delta t]$

→ The equation above relates $\partial_t P$, $\partial_{ss}P$ and $E[(R_{t,t+\Delta t})^2 \mid \mathcal{F}_t]$

Let us <u>assume</u> that

$$E\left[(R_{t,t+\Delta t})^2 \mid \mathcal{F}_t \right] = \text{const} = \sigma^2 \Delta t \quad , \forall t$$

where $\sigma > 0$ is a parameter describing the <u>annualized realized volatility</u> of S

↳ $\sigma = 20\% = 0.2$ means $0.2 \dfrac{1}{\sqrt{\text{years}}}$

↳ daily volatility is $\dfrac{\sigma}{\sqrt{250}}$

(because 1 year $\approx$ 250 trading day)

→ Then, we get the condition

$$-\partial_t P(t, S_t)\cancel{\Delta t} - \frac{1}{2}\partial_{ss} P(t, S_t)\, S_t^2 \sigma^2 \cancel{\Delta t} = 0$$

which is satisfied if the function P solves the equation

$$\begin{cases} \partial_t P(t,s) + \frac{1}{2}\sigma^2 s^2 \partial_{ss} P(t,s) = 0 & \forall t \in [0,T),\ \forall s > 0 \\ P(T,s) = g(s) & \forall s > 0 \end{cases}$$

This is the Black-Scholes Partial Differential Equation (PDE)

or the _____ pricing equation

with volatility parameter σ, interest rate $r = 0$,

payoff g.

→ Solution of the PDE : our choice of the function P

→ Once we are able to evaluate the function P,

the delta-hedge position is

$$\delta_t = \partial_s P(t, S_t)$$

solution of the PDE

observed on the market

→ Computational problem : solve the PDE.

▸ With the choice above for the function P, we get:

$$P\&L_{t,t+\Delta t} = \underbrace{-\partial_t P(t,S_t)\Delta t}_{\substack{\frac{1}{2}\sigma^2 S_t^2\, \partial_{ss} P(t,S_t) \\ \text{because of the PDE}}} - \frac{1}{2}\partial_{ss} P(t,S_t) S_t^2 \left(\frac{\Delta S_t}{S_t}\right)^2$$

$$= \frac{1}{2}\underbrace{\partial_{ss} P(t,S_t)\, S_t^2}_{\text{"cash gamma"}} \left(\sigma^2 \Delta t - \left(\frac{\Delta S_t}{S_t}\right)^2\right) \quad (*)$$

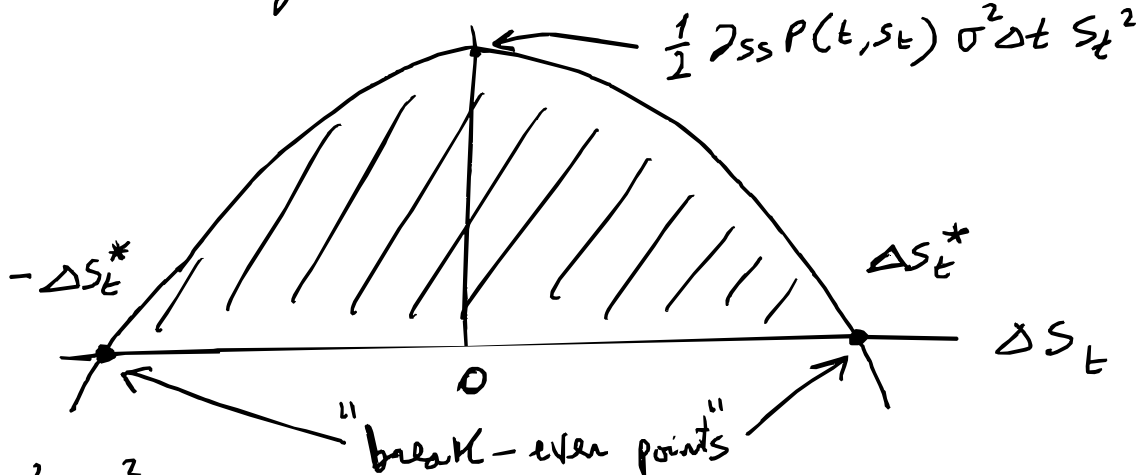
The approximation $(*)$ is called the "<u>theta</u> - <u>gamma</u>" approximation of the P&L

↳ of order 1 in Δt and order 2 in ΔS

- The theta-gamma P&L of the delta-hedged option depends on:
  - the option's gamma $\partial_{ss} P(t,S_t)$
  - the realized square increment ΔS_t, which is random.
  - the parameter σ.

- If $\partial_{ss} P(t,s) \geq 0 \quad \forall t, \forall s$

then the theta-gamma PnL behaves as follows



$$(\Delta S_t^*)^2 = S_t^2 \sigma^2 \Delta t$$

$$|\Delta S_t^*| = S_t \sigma \sqrt{\Delta t}$$

- If $|\Delta S_t| < |\Delta S_t^*|$, then $PnL_{t,t+\Delta t} > 0$
- If $|\Delta S_t| > |\Delta S_t^*|$, then $PnL_{t,t+\Delta t} < 0$

- <u>Case $r \neq 0$</u>

Recall that

$$PnL_{t,t+\Delta} = r(V_t - P(t,S_t))\Delta t - \Theta(t,S_t) - \frac{1}{2}\partial_{SS}P(t,S_t)(\Delta S_t)^2 \qquad (Eq\ 1)$$

We introduce :

$$\widetilde{PnL}_{t_i,t_i+1} = \frac{\text{position at } t_{i+1}}{(1+r\Delta t)^{i+1}} - \frac{\text{position at } t_i}{(1+r\Delta t)^i}$$

<u>discounted PnL</u>

<u>at time t_0</u>

$$= \frac{-P(t_{i+1},S_{t_{i+1}}) + V_{t_i}}{(1+r\Delta t)^{i+1}} - \frac{-P(t_i,S_{t_i}) + V_{t_i}}{(1+r\Delta t)^i}$$

We consider

- $t_{i+1} - t_i = \Delta t, \ \forall i$
- $(1+r\Delta t)^{-i}$ is the discount factor

By injecting (Eq 1) into $\widetilde{PnL}_{t_i,t_{i+1}}$, we get:

$$\widetilde{PnL}_{t_i,t_{i+1}} = \frac{1}{(1+r\Delta t)^{i+1}} \left[-\theta(t_i, S_{t_i})\Delta t - \frac{1}{2}\partial_{ss}P(t_i, S_{t_i})(S_{t_i})^2 (\Delta S_i)^2 \right]$$

where $\Delta S_i = \Delta S_{t_i} = S_{t_{i+1}} - S_{t_i}$

- Same argument: if we impose

$$E[\widetilde{PnL}_{t_i,t_{i+1}} | \mathcal{F}_{t_i}] = 0$$

and we assume that $E[(R_{t_i,t_{i+1}})^2 | \mathcal{F}_{t_i}] = \sigma^2 \Delta t, \forall i$

then we get the condition

$$-\theta(t_i, S_{t_i})\cancel{\Delta t} - \frac{1}{2}\partial_{ss}P(t_i, S_{t_i}) S_{t_i}^2 \sigma^2 \cancel{\Delta t} = 0,$$

which is satisfied if P satisfies the following PDE:

$\hookrightarrow$ using $\theta(t,S) = \partial_t P(t,S) + rS\partial_S P(t,S) - rP(t,S)$

$$\text{PDE}_{BS} \quad \begin{cases} \partial_t P(t,s) + r s \partial_s P(t,s) + \frac{1}{2}\sigma^2 s^2 \partial_{ss} P(t,s) - r P(t,s) = 0 & \forall t \in [0,T),\ \forall x > 0 \\ P(T,s) = g(s) & \forall s \end{cases}$$

↳ This is the <u>Black-Scholes pricing PDE</u>

with payoff g, volatility parameter σ, interest rate r.

▷ Note: in the (PDEBS), t and s are variables (no S_t)

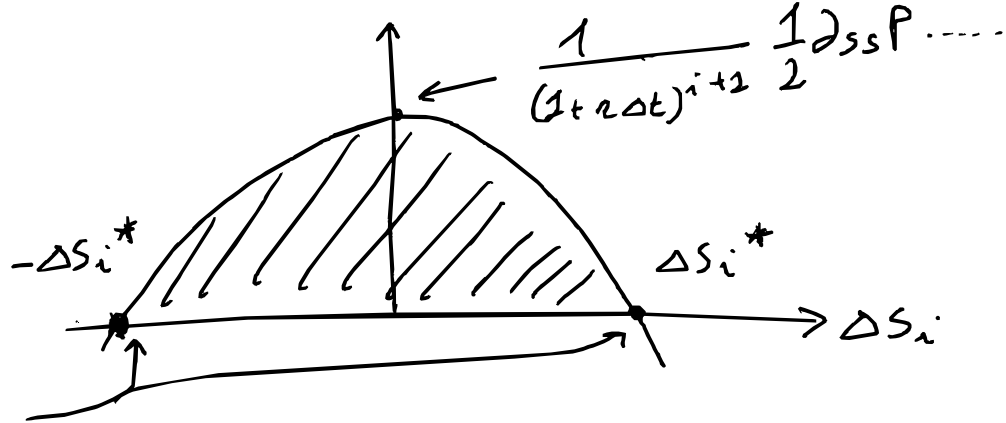
$t \in [0,T)$

$s \in \mathbb{R}_+$

- The resulting theta-gamma approximation of the PnL is:

$$PnL_{t_i,t_{i+1}} = \frac{1}{(1+r\Delta t)^{i+1}} \frac{1}{2} \partial_{ss} P(t_i, S_{t_i}) (S_{t_i})^2 \left(\sigma^2 \Delta t - \left(\frac{\Delta S_i}{S_{t_i}} \right)^2 \right)$$

because of the (PDE BS)



break-even points

$$|\Delta S_i^*| = S_{t_i} \, \sigma \sqrt{\Delta t}$$

are still the same (they do not depend on r).

Of course, the global PnL over the lifetime of the option is:

$$\widetilde{PnL}_{0,T} = \sum_{i=0}^{n-1} \widetilde{PnL}_{t_i, t_{i+1}}$$

$$\approx \sum_{i=0}^{n-1} \frac{1}{(1+r\Delta t)^{i+1}} \frac{1}{2} \partial_{SS} P(t_i, S_{t_i}) S_{t_i}^2 \left(\sigma^2 \Delta t - \left(\frac{\Delta S_i}{S_{t_i}} \right)^2 \right)$$